Mathematical Foundations of the Real-Time Market Data and Volatility Engine

Technical Documentation and Parameter Guide for `market_data_v2.py`

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Abstract

This document provides an accessible, step-by-step mathematical breakdown of the market data and volatility estimation pipeline implemented in `market_data_v2.py`. Designed to feed quantitative risk and liquidation engines with clean empirical inputs, the module resolves several practical challenges unique to digital assets. We explain: why 24/7 continuous trading invalidates traditional overnight gap models; how forward-looking implied volatility is retrieved from Deribit's DVOL index; the mathematics of the Parkinson (1980) and Rogers-Satchell (1991) continuous range estimators; the composite Yang-Zhang (2000) estimator; rolling multi-period funding rate smoothing; and Bayesian shrinkage regularizers to tame Merton's standard error in drift estimation. Every section provides plain English intuition, explicit numbered formulas, parameter references, and practical limitations.

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1 The Reality of 24/7 Crypto Markets vs. Traditional Finance

1.1 Simple Language Explanation

In traditional stock exchanges and regulated futures markets (such as the Chicago Mercantile Exchange, CME), trading pauses on Friday afternoon and reopens on Sunday evening. Over the weekend, macro events unfold while trading is paused. When the market reopens, the opening price often jumps far away from Friday's closing price. This is known as a weekend price gap.

In cryptocurrency derivative markets (such as Binance, Bybit, and Deribit), trading never pauses. It operates 24 hours a day, 7 days a week, 365 days a year. A new daily candle opens in the exact same millisecond that the previous daily candle closes.

1.2 Mathematical Consequence for Volatility Measurement

Because crypto markets trade continuously, the opening price of the current period O_t is virtually identical to the closing price of the previous period C_{t-1}:

$$O_t \approx C_{t-1} \implies u_t = \ln\left(\frac{O_t}{C_{t-1}}\right) \approx 0 \tag{1}$$

As a result, the overnight return variance $\sigma^2_{\text{open}} \approx 0$. Traditional equity volatility models that rely heavily on overnight gap variance become redundant. For 24/7 continuous crypto markets, high-low range estimators (such as Parkinson and Rogers-Satchell) provide much more efficient and reliable volatility estimates.

2 Principle 1: Deribit Forward-Looking Implied Volatility Index (DVOL)

2.1 Simple Language Explanation

Implied Volatility (IV) does not look backward at historical price prints; it looks forward at what institutional option traders expect over the next 30 days. The Deribit DVOL index extracts this collective market expectation directly from live option order books.

- **Whitelist Fast-Fail Guard:** Deribit only maintains liquid, high-volume options markets for major cryptocurrencies: Bitcoin (BTC), Ethereum (ETH), and Solana (SOL). The engine checks this whitelist first. If a user queries an altcoin, the engine skips the network call immediately, avoiding unnecessary HTTP timeouts and falling back directly to historical estimators.

2.2 Mathematical Formulation

The DVOL index is calculated using a model-free implied variance formula over a 30-day forward horizon ($T = 30/365$ years):

$$\sigma^2_{\text{IV}} = \frac{2e^{rT}}{T}\sum_i \frac{\Delta K_i}{K_i^2} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 \tag{2}$$

where $Q(K_i)$ is the mid-market price of the out-of-the-money option strike K_i, F is the forward index price, and K_0 is the strike just below F. The index is quoted as an annualized

percentage point. The function `fetch_deribit_dvol` converts it to a standard decimal ratio:

$$\sigma_{\text{DVOL}} = \frac{\text{DVOL}_{\text{close}}}{100.0} \tag{3}$$

2.3 Parameter Reference

- `currency` (`str`): Base cryptocurrency asset symbol (e.g., `"BTC"`, `"ETH"`, `"SOL"`).
- `DERIBIT_SUPPORTED_DVOL_CURRENCIES`: Set of supported assets (`{"BTC", "ETH", "SOL"}`).

2.4 Limitations

- **Unavailable for Altcoins:** Liquid option order books do not exist for the vast majority of altcoins, causing `fetch_deribit_dvol` to return `None`.
- **Fixed 30-Day Window:** The DVOL index is calibrated to a fixed 30-day constant maturity and may not reflect very short-term intraday volatility spikes.

3 Principle 2: Continuous Range Estimators (Parkinson and Rogers-Satchell)

3.1 Simple Language Explanation

- **Parkinson Volatility (1980):** Instead of looking only at closing prices, Parkinson evaluates the extreme price spread between the High (H_t) and Low (L_t) of each candle. A wide spread indicates turbulent intraday trading. Statistically, it is roughly five times more efficient than classical close-to-close sample standard deviation.
- **Rogers-Satchell Volatility (1991):** Parkinson has one structural flaw: if the market is trending strongly upward or downward, the high-low spread expands, leading Parkinson to mistake strong directional trend for random volatility. Rogers-Satchell resolves this by incorporating Open, High, Low, and Close prices simultaneously, remaining unbiased even in the presence of non-zero drift.

3.2 Mathematical Formulation

3.2.1 1. Parkinson (1980) High-Low Estimator

Using the properties of the running extremum of a continuous Brownian motion:

$$\sigma_P^2 = \frac{1}{4\ln(2) \cdot n} \sum_{t=1}^{n} \left[\ln\left(\frac{H_t}{L_t}\right) \right]^2 \tag{4}$$

Annualizing across continuous crypto periods ($N_{\text{trading}} = 365.0$):

$$\sigma_{P,\text{annual}} = \sqrt{\sigma_P^2 \times N_{\text{trading}}} \tag{5}$$

3.2.2 2. Rogers-Satchell (1991) Drift-Independent Estimator

Measuring intraday price dispersion without upward bias from non-zero drift ($\mu \neq 0$):

$$\sigma_{\text{RS}}^2 = \frac{1}{n}\sum_{t=1}^{n}\left[\ln\left(\frac{H_t}{C_t}\right)\ln\left(\frac{H_t}{O_t}\right) + \ln\left(\frac{L_t}{C_t}\right)\ln\left(\frac{L_t}{O_t}\right)\right] \tag{6}$$

$$\sigma_{\text{RS,annual}} = \sqrt{\max\left(0.0,\ \sigma_{\text{RS}}^2\right) \times N_{\text{trading}}} \tag{7}$$

3.3 Parameter Reference

- `highs`, `lows`, `opens`, `closes` (`np.ndarray`): Historical price vectors.
- `trading_periods` (`float`): Annualization scale factor (default `365.0` for continuous crypto).

3.4 Limitations

- **Parkinson Trend Bias:** Parkinson systematically overestimates volatility during strong, smooth directional rallies. Rogers-Satchell is preferred in trending regimes.
- **Sensitivity to Bad Ticks:** A single erroneous exchange wick (a flash crash on an illiquid order book) inflates high-low estimators for the entire sample window.

4 Principle 3: Composite Yang-Zhang (2000) Estimator

4.1 Simple Language Explanation

The Yang-Zhang Estimator is a comprehensive historical volatility metric combining three separate sources of information: overnight jump variance, open-to-close daytime variance, and continuous Rogers-Satchell range variance. It blends these components using an optimal weighting parameter $k(n)$ mathematically chosen to minimize total estimation variance.

4.2 Mathematical Formulation

Given $N \geq 3$ historical candles indexed $t \in \{0, 1, \ldots, N-1\}$, let $n = N - 1$ denote the number of lagged return intervals. For each $t \in \{1, \ldots, n\}$:

$$u_t = \ln\left(\frac{O_t}{C_{t-1}}\right), \qquad \sigma_{\text{open}}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(u_t - \bar{u})^2 \tag{8}$$

$$c_t = \ln\left(\frac{C_t}{O_t}\right), \qquad \sigma_{\text{close}}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(c_t - \bar{c})^2 \tag{9}$$

The Rogers-Satchell component calculated across the lagged slice $t \in \{1, \ldots, n\}$ is:

$$\sigma_{\text{RS}}^2 = \frac{1}{n}\sum_{t=1}^{n}\left[\ln\left(\frac{H_t}{C_t}\right)\ln\left(\frac{H_t}{O_t}\right) + \ln\left(\frac{L_t}{C_t}\right)\ln\left(\frac{L_t}{O_t}\right)\right] \tag{10}$$

The optimal minimum-variance weighting parameter $k(n)$ is:

$$k(n) = \frac{0.34}{1.34 + \frac{n+1}{n-1}} \tag{11}$$

The combined variance and annualized volatility are:

$$\sigma_{\text{YZ}}^2 = \sigma_{\text{open}}^2 + k \cdot \sigma_{\text{close}}^2 + (1-k) \cdot \sigma_{\text{RS}}^2 \tag{12}$$

$$\sigma_{\text{YZ,annual}} = \sqrt{\max\left(0.0,\ \sigma_{\text{YZ}}^2\right) \times N_{\text{trading}}} \tag{13}$$

4.3 Parameter Reference

- `opens`, `highs`, `lows`, `closes`: Price series arrays with length ≥ 3.
- `trading_periods`: Annual trading periods (`365.0`).

5 Principle 4: Rolling Multi-Period Funding Rate Smoothing

5.1 Simple Language Explanation

In perpetual futures, cash transfers occur between longs and shorts every 8 hours. During short squeezes, a single 8-hour funding fee can temporarily spike to $+0.1\%$ or $+0.2\%$.

If you take that single spike and multiply it by $1{,}095$ (3×365 periods) to annualize it, the annualized funding rate appears to be **+110% to +220%**! If a Monte Carlo ruin simulator uses this annualized spike, it will falsely project that the trader's collateral will be wiped out within days.

The function `fetch_rolling_funding_rate` resolves this by downloading the past M funding intervals (default $M = 21$, or 7 full days) and computing their smoothed rolling average. This filters out temporary noise and accurately captures the sustainable long-term cash drag.

5.2 Mathematical Formulation

The function retrieves the historical sequence of the last M funding periods (default $M = 21$) and calculates the arithmetic mean:

$$\bar{r}_{\text{8h}} = \frac{1}{M} \sum_{m=1}^{M} r_{\text{8h}}^{(m)} \tag{14}$$

The annualized smoothed funding rate is computed by scaling over all $1{,}095$ eight-hour periods in a calendar year:

$$r_{\text{annual,smoothed}} = \bar{r}_{\text{8h}} \times 3.0 \times 365.0 = 1095 \cdot \bar{r}_{\text{8h}} \tag{15}$$

The instantaneous spot rate is also retained for comparative monitoring:

$$r_{\text{annual,spot}} = r_{\text{spot,8h}} \times 1095.0 \tag{16}$$

5.3 Parameter Reference

- `exchange`: CCXT exchange connection instance initialized with `defaultType: "swap"`.
- `symbol`: Perpetual swap market pair (e.g., `"BTC/USDT"`).
- `lookback_periods` (M): Number of historical 8-hour funding periods (default $21 = 7$ days).
- `default_fallback`: Fallback rate if the exchange API fails (0.01% per 8h).

5.4 Limitations

- **Lag During Structural Trend Shifts:** A 7-day rolling average protects against single-period spikes, but if the market enters a sustained multi-month regime change with consistently elevated funding, the rolling average will take several days to fully adapt.

6 Principle 5: Merton's Drift Estimation Trap and Bayesian Shrinkage

6.1 Simple Language Explanation

If you observe Bitcoin over the past 90 days and calculate an average growth rate of $+40$% per year, can you safely assume it will continue to grow at $+40$% per year?

In 1980, Nobel laureate Robert Merton proved mathematically that estimating the average drift (return) of a volatile asset from a short sample contains massive statistical error—often exceeding ± 140%! If a risk engine feeds this raw, noisy drift estimate into a leverage formula like the Kelly criterion, the formula will recommend dangerously high leverage right before a reversal.

The engine uses **Bayesian Shrinkage** to pull noisy historical drift estimates conservatively toward a neutral baseline (0%), preventing risk engines from overcommitting capital.

6.2 Mathematical Formulation

The naive sample annualized log drift from $N-1$ daily close returns is:

$$\hat{\mu}_{\text{naive}} = \left(\frac{1}{N-1} \sum_{t=1}^{N-1} \ln\left(\frac{C_t}{C_{t-1}} \right) \right) \times 365.0 \tag{17}$$

Merton (1980) proved that the standard error of drift estimation depends strictly on volatility σ and total observation time in years T_{years}:

$$\text{SE}(\hat{\mu}) = \frac{\sigma}{\sqrt{T_{\text{years}}}} \tag{18}$$

To tame this error, we establish a conservative prior distribution centered at zero ($\mu_{\text{prior}} = 0.0$) with prior variance $\sigma^2_{\text{prior}} = (0.15)^2$. The Bayesian posterior shrinkage weight w is:

$$w = \frac{\sigma^2_{\text{prior}}}{\sigma^2_{\text{prior}} + \text{SE}(\hat{\mu})^2} \tag{19}$$

The regularized annualized drift used by the engine is:

$$\mu_{\text{shrunk}} = w \cdot \hat{\mu}_{\text{naive}} + (1-w) \cdot \mu_{\text{prior}} = w \cdot \hat{\mu}_{\text{naive}} \tag{20}$$

6.3 Parameter Reference

- `closes`: Historical close price vector.
- `sigma`: Active annualized volatility.
- `trading_periods`: Annual trading periods (`365.0`).
- `prior_mean`: Baseline long-term drift prior (default `0.0`).
- `prior_std`: Confidence in the prior (default $0.15 = 15$%).

7 Principle 6: Active Volatility Selection Hierarchy and Exchange Failover

7.1 Simple Language Explanation

The primary extraction function `get_market_data` is built to be resilient:

1. **Intelligent Volatility Hierarchy:** It prioritizes forward-looking option volatility (DVOL) whenever available. If an altcoin is queried (or Deribit is unreachable), it falls back seamlessly to the Yang-Zhang realized range estimator.
2. **Automatic Exchange Failover:** The engine first queries Binance linear swap endpoints. If Binance is experiencing outages or rate limits, the client automatically switches to Bybit without crashing the application.

7.2 Mathematical Formulation

The active volatility metric σ_{active} supplied to the liquidation engine is selected via:

$$\sigma_{\text{active}} = \begin{cases} \sigma_{\text{DVOL}} & \text{if Deribit DVOL query succeeds and } \sigma_{\text{DVOL}} > 0 \\ \sigma_{\text{YZ,annual}} & \text{otherwise (fallback to Yang-Zhang)} \end{cases} \tag{21}$$

8 Master Parameter Catalog for the Market Data Engine

Table 1 details all inputs, metrics, and parameters in `market_data_v2.py`.

9 Summary and System Integration

The outputs of `market_data_v2.py` provide the empirical parameters required by the liquidation ruin engine:

1. **Prevents False Liquidations:** Supplying `annualized_funding_rate` (Equation 15) protects Monte Carlo simulations from spurious early liquidations caused by temporary funding rate spikes.
2. **Guards Leverage Optimization:** Supplying `annualized_drift` (Equation 20) prevents the Kelly criterion from recommending unsafe leverage based on noisy short-term trends.
3. **Captures True Dispersion:** Implementing Parkinson and Rogers-Satchell estimators ensures volatility measurements reflect continuous 24/7 price action without depending on non-existent overnight gaps.

Table 1: Market Data Engine Parameter Catalog

Variable Name	Symbol	Type	Default	Operational Role and Financial Meaning
`symbol`	—	`str`	`"BTC/USDT"`	Target perpetual swap pair.
`currency`	—	`str`	`"BTC"`	Base cryptocurrency ticker used for DVOL query.
`timeframe`	—	`str`	`"1d"`	Candlestick resolution (daily).
`limit`	N	`int`	90	Number of historical candles sampled.
`trading_periods`	N_{tr}	`float`	365.0	Annualization scaling factor for continuous crypto.
`lookback_periods`	M	`int`	21	Number of 8-hour periods in rolling funding window.
`prior_std`	σ_{prior}	`float`	0.15	Prior uncertainty on drift used for Bayesian shrinkage.
`parkinson_volatility`	σ_P	`float`	—	Annualized Parkinson high-low volatility.
`rogers_satchell_volatility`	σ_{RS}	`float`	—	Annualized Rogers-Satchell drift-independent volatility.
`yang_zhang_volatility`	σ_{YZ}	`float`	—	Annualized minimum-variance Yang-Zhang volatility.
`implied_volatility`	σ_{IV}	`float`	—	Deribit forward-looking 30-day DVOL volatility.
`active_volatility`	σ_{act}	`float`	—	Active volatility selected via Equation (21).
`smoothed_funding_rate_8h`	$\bar{r}_{8\text{h}}$	`float`	—	7-day rolling average 8-hour funding rate.
`annualized_funding_rate`	r_{ann}	`float`	—	Smoothed annualized funding rate fed to ruin engine.
`annualized_drift`	μ_{shrunk}	`float`	—	Bayesian-regularized annualized drift.
`merton_standard_error`	SE	`float`	—	Merton statistical standard error of sample drift.